Unit 3 (Part 8) – Fractions (× and ÷)

Multiplication and Division of Algebraic Fractions

Points to remember when multiplying & dividing algebraic fractions:

1. When dividing remember to "**TIP and TIMES**". The divide symbol affects the factor/fraction that is directly after it.

   e.g. $\frac{1}{2} \div \mathbf{\frac{3}{5}} \times \frac{3}{4} = \frac{1}{2} \mathbf{\times \frac{5}{3}} \times \frac{3}{4} = \frac{1 \cdot 5 \cdot 3}{2 \cdot 3 \cdot 4}$

   "Top × top; answer on top.
   Bottom × bottom; answer at the bottom."

2. Fully factorise all numerators and denominators to identify common factors.

3. Consider the sign change rule.

   1. $(y - x) = -(x - y)$
   2. $(b + a)(d + c) = (a + b)(c + d)$
   3. $(a + b)(d - c) = -(a + b)(c - d)$
   4. $(b - a)(d - c) = (a - b)(c - d)$
   5. $(b - a)^2 = (a - b)^2$

4. Divide common factors and *simplify*.
   To *simplify* a fraction means to divide away the HCF of the numerator and denominator.

Simplify the following:

1. $\dfrac{4a - 4b}{2b - 2a}$

2. $\dfrac{3x^2 - 27}{x^2 - 4x - 5} \div \dfrac{x^2 + 4x + 3}{2x - 10} \times \dfrac{x^2 + 2x + 1}{6}$

3. $\dfrac{a-4}{8-2a}$

4. $\dfrac{x^2-x-6}{3x-9} \div \dfrac{2x^2-8}{5x-10}$

5. $\dfrac{5a-5ab}{2b^2+b-3} \times \dfrac{6-4b}{5b^2+10b+5} \div \dfrac{2a+4ab}{2b^2+3b+1}$

6. $\frac{x^2 - 2x - 8}{8 - 2x} \times \frac{3x - 6}{x^3 + 8} \div \frac{2a - xa}{3x^2 - 6x + 12}$

7. $\frac{a + 2 - 3a^2 - 6a}{a + 2}$

8. $\frac{2(a + 1)^2 + (a + 1)}{a + 1}$

Homework: Ex. 13 pg. 34 – 35 (Mind Action Series new edition, 2022)